The re-entrancy obstruction is the token-mutation bit: an analytic proof for orchestration Zappa–Szép products

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Abstract

We give a fully analytic proof — no enumeration of factorization systems, no machine-verified isomorphism — of the re-entrancy obstruction theorem for supervisor–worker agent orchestration. Model the handoff topology as a small category $\mathcal{K}_\varepsilon$ carrying a one-bit turn token, parametrized by a single bit $\varepsilon \in \mathbb{Z}/2$: the worker's re-entrant outcome sends the dispatch p to $s_2 \circ p = q \cdot \tau^\varepsilon$, so $\varepsilon = 0$ means the worker *fixes* the supervisor's token and $\varepsilon = 1$ means it *mutates* it. Fixing the token-internal moves $\mathcal{D}$ as the right factor, we prove

$$[\omega(\mathcal{K}_\varepsilon)] \;=\; \varepsilon \cdot (\text{generator}) \quad \text{in} \quad H^2(\mathrm{Sk}_\mathcal{C}; \mathcal{D}) \cong \mathbb{Z}/2.$$

That is, *the degree-two obstruction class equals the token-mutation bit, on the nose.* It follows that the joint orchestration admits a Zappa–Szép product $\mathcal{K}_\varepsilon = \mathcal{C} \bowtie \mathcal{D}$ (equivalently a distributive law $\lambda : \mathcal{DC} \Rightarrow \mathcal{CD}$) if and only if $\varepsilon = 0$, i.e. if and only if the worker fixes the token; and when $\varepsilon = 1$ the obstruction is the nonzero generator of $H^2 \cong \mathbb{Z}/2$, the unprotected-re-entrancy failure mode. The proof assembles two cited theorems — the pairwise Zappa–Szép criterion and the (G)-obstruction-is-H^2 classification — with a direct orbit and cochain computation on the parametrized family; no new cohomology is developed. This promotes the orchestration node from COMPUTED to PROVED.

1 Statement and cited scaffolding

Two theorems are used as black boxes; both are proved in companion notes and are cited, never reproved.

(T2) *Pairwise Zappa–Szép criterion* [1]. For a small category $\mathcal{K}$ and a wide subcategory $\mathcal{D} \subseteq \mathcal{K}$, there exists a wide $\mathcal{C} \subseteq \mathcal{K}$ with $(\mathcal{C}, \mathcal{D})$ a strict factorization system — equivalently a distributive law $\lambda : \mathcal{DC} \Rightarrow \mathcal{CD}$ satisfying the matched-pair axioms ZS1–ZS4, equivalently $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ — if and only if

- (L) each hom-presheaf $\mathrm{Hom}_\mathcal{K}(-, b)$ is free as a right $\mathcal{D}$-set; and
- (G) free bases can be chosen closing into a wide subcategory.

(T3) *The (G)-obstruction is an H^2 class* [2]. Assume (L) and the hypothesis (H) below (abelian vertex groups, trivial left action). Then the free $\mathcal{D}$-orbits form a category $\mathrm{Sk}_\mathcal{C}$, the vertex automorphism groups form a presheaf $\mathcal{D} : \mathrm{Sk}_\mathcal{C}^{\mathrm{op}} \to \mathbf{Ab}$, every transversal T carries a defect 2-cocycle ω_T, regauging changes it by a coboundary, and

$$\text{(G) holds} \iff [\omega] := [\omega_T] = 0 \in H^2(\mathrm{Sk}_\mathcal{C}; \mathcal{D}).$$

The cohomology itself is classical (Rosebrugh–Wood [4] for existence, Baues–Wirsching [5] for the abelian classification); it is cited, not reproved.

Hypothesis (H) [2]. *(i)* $\mathcal{D}$ is a disjoint union of vertex groups $G_a = \mathrm{Aut}_{\mathcal{D}}(a)$, each abelian, with $\mathcal{D}(a, b) = \varnothing$ for $a \neq b$. *(ii)* For every generator $c : a \to b$ and every $\psi \in G_b$, $\psi \circ c$ lies in the $\mathcal{D}$-orbit of c (trivial left action).

(We refer to this displayed hypothesis as (H).)

The reading "an agent's interface is a polynomial functor / container, its dynamics a coalgebra" is Spivak's prior art and is not claimed. The contribution here is the analytic identification of the re-entrancy obstruction with an explicit degree-two class, uniformly across the token-mutation parameter.

2 The parametrized orchestration category

Fix three handoff roles: S (supervisor), W (worker running), R (result/return point). The supervisor carries a one-bit *turn token*; toggling it is an involution τ with $\tau^2 = \mathrm{id}_{\mathsf{S}}$, so $\mathrm{End}(\mathsf{S}) = \{\mathrm{id}_{\mathsf{S}}, \tau\} \cong \mathbb{Z}/2$.

Definition 1 (The family $\mathcal{K}_\varepsilon$)**.** For $\varepsilon \in \mathbb{Z}/2$, let $\mathcal{K}_\varepsilon$ be the small category with objects $\mathsf{S}, \mathsf{W}, \mathsf{R}$ and morphisms

$$\begin{aligned}&\mathrm{End}(\mathsf{S}) = \{\mathrm{id}_{\mathsf{S}}, \tau\}, \quad \mathrm{End}(\mathsf{W}) = \{\mathrm{id}_{\mathsf{W}}\}, \quad \mathrm{End}(\mathsf{R}) = \{\mathrm{id}_{\mathsf{R}}\},\\ &\mathrm{Hom}(\mathsf{S}, \mathsf{W}) = \{p,\ p\tau\}, \quad \mathrm{Hom}(\mathsf{S}, \mathsf{R}) = \{q,\ q\tau\}, \quad \mathrm{Hom}(\mathsf{W}, \mathsf{R}) = \{s,\ s_2\},\end{aligned}$$

and no morphisms $\mathsf{W} \to \mathsf{S}$, $\mathsf{R} \to \mathsf{S}$, $\mathsf{R} \to \mathsf{W}$ (the topology is acyclic: $\mathsf{S} \to \mathsf{W} \to \mathsf{R}$ with a direct $\mathsf{S} \to \mathsf{R}$). Here $p\tau := p \circ \tau$ and $q\tau := q \circ \tau$, and $p \neq p\tau$, $q \neq q\tau$ by construction (the token genuinely distinguishes the two dispatch states). Composition through the middle is

$$s \circ p = q, \qquad s_2 \circ p = q \cdot \tau^{\varepsilon},$$

extended by right τ-equivariance: $s \circ p\tau = q\tau$, $s_2 \circ p\tau = q \cdot \tau^{\varepsilon+1}$; identities and $\tau^2 = \mathrm{id}_{\mathsf{S}}$ as stated.

The single bit ε is the entire content: s_2 is the re-entrant worker outcome, and $\varepsilon = [\, s_2 \circ p = q\tau \,]$ records whether that outcome *mutates* the token ($\varepsilon = 1$) or *fixes* it ($\varepsilon = 0$). We name the two instances $\mathcal{K}_{\mathrm{ok}} := \mathcal{K}_0$ (state-protected re-entry, a lock) and $\mathcal{K}_{\mathrm{bug}} := \mathcal{K}_1$ (unprotected re-entry).

Lemma 2 ($\mathcal{K}_\varepsilon$ is a category)**.** *For each $\varepsilon \in \mathbb{Z}/2$, $\mathcal{K}_\varepsilon$ is a well-defined small category.*

Proof. Every non-identity morphism raises the poset level $\mathsf{S} < \mathsf{W} < \mathsf{R}$ except the S-endomorphism τ, so the only composites to specify are (a) right multiplication by τ on S-sourced arrows, which is a group action ($\tau^2 = \mathrm{id}$, and $f \cdot \tau^2 = f$), and (b) the two length-two composites $s \circ p,\ s_2 \circ p : \mathsf{S} \to \mathsf{R}$, given above; all longer composites factor through R (a sink) or begin with an S-endomorphism. Associativity therefore reduces to compatibility of (a) and (b): for $f \in \{s, s_2\}$,

$$(f \circ p) \circ \tau \ = \ f \circ (p \circ \tau) \ = \ f \circ p\tau,$$

which holds *by definition* of $f \circ p\tau$ as $(f \circ p) \cdot \tau$; and $(f \circ p) \circ \tau^2 = f \circ p$ since $\tau^2 = \mathrm{id}_{\mathsf{S}}$. Unit laws are immediate. Hence $\mathcal{K}_\varepsilon$ is a category. (Machine-confirmed for both ε in `orchestration_zs_parametrized.py`.) □

We fix once and for all the wide right factor of token-internal ("reader") moves,

$$\mathcal{D} = \{\mathrm{id}_{\mathsf{S}},\ \tau,\ \mathrm{id}_{\mathsf{W}},\ \mathrm{id}_{\mathsf{R}}\} \subseteq \mathcal{K}_\varepsilon,$$

a wide subcategory (it contains all identities and is closed: $\tau \circ \tau = \mathrm{id}_{\mathsf{S}}$). Its vertex groups are $G_{\mathsf{S}} = \{\mathrm{id}_{\mathsf{S}}, \tau\} \cong \mathbb{Z}/2$ and $G_{\mathsf{W}} = G_{\mathsf{R}} = 0$.

3 Freeness and the hypothesis hold, uniformly in ε

Lemma 3 ((L) holds)**.** *For each ε, every hom-presheaf $\mathrm{Hom}_{\mathcal{K}_\varepsilon}(-,b)$ is a free right $\mathcal{D}$-set.*

Proof. The right $\mathcal{D}$-action is precomposition; since $\mathcal{D}(a,b)=\varnothing$ for $a\neq b$ and the only nonidentity element is $\tau\in G_{\mathsf{S}}$, the action is trivial on W- and R-sourced arrows and is the regular $\mathbb{Z}/2$-action, by right multiplication by τ, on S-sourced arrows. Decompose each hom-presheaf into $\mathcal{D}$-orbits:

$$\begin{aligned}
\mathrm{Hom}(-,\mathsf{S}) &= \{\mathrm{id}_{\mathsf{S}},\tau\} && \text{(one regular } \mathbb{Z}/2\text{-orbit)},\\
\mathrm{Hom}(-,\mathsf{W}) &= \underbrace{\{p,p\tau\}}_{\text{apex } \mathsf{S}} \sqcup \underbrace{\{\mathrm{id}_{\mathsf{W}}\}}_{\text{apex } \mathsf{W}},\\
\mathrm{Hom}(-,\mathsf{R}) &= \underbrace{\{q,q\tau\}}_{\text{apex } \mathsf{S}} \sqcup \underbrace{\{s\}}_{\text{apex } \mathsf{W}} \sqcup \underbrace{\{s_2\}}_{\text{apex } \mathsf{W}} \sqcup \underbrace{\{\mathrm{id}_{\mathsf{R}}\}}_{\text{apex } \mathsf{R}}.
\end{aligned}$$

Each S-apex orbit has two distinct elements permuted freely by τ ($p\neq p\tau$, $q\neq q\tau$, $\mathrm{id}_{\mathsf{S}}\neq\tau$), hence is a free G_{S}-torsor $\cong\mathcal{D}(-,\mathsf{S})$; each W- or R-apex orbit is a singleton with trivial group, $\cong\mathcal{D}(-,\mathsf{W})$ resp. $\mathcal{D}(-,\mathsf{R})$. So every hom-presheaf is a coproduct of representable $\mathcal{D}$-sets, i.e. free. This uses only that $p\neq p\tau$ and $q\neq q\tau$; it is independent of ε (the value of $s_2\circ p$ lies inside the orbit $\{q,q\tau\}$ either way). ☐

Lemma 4 ((H) holds)**.** *For each ε, $\mathcal{K}_\varepsilon$ with $\mathcal{D}$ satisfies Hypothesis (H).*

Proof. *(i)* $\mathcal{D}$ is the disjoint union of the vertex groups $G_{\mathsf{S}}\cong\mathbb{Z}/2$ (abelian), $G_{\mathsf{W}}=G_{\mathsf{R}}=0$; there are no cross arrows. *(ii)* The left action is postcomposition by a target vertex automorphism $\psi\in G_b$. The only nontrivial vertex group sits at S, and S is a *source*: no non-identity generator has codomain S (nothing maps into S but its own endomorphisms). Thus for every generator $c:a\to b$ with $b\in\{\mathsf{W},\mathsf{R}\}$ the group G_b is trivial and $\psi\circ c=c\in\mathrm{orb}(c)$; the remaining case is the regular orbit at S, where $\tau\circ\mathrm{id}_{\mathsf{S}}=\tau=\mathrm{id}_{\mathsf{S}}\circ\tau\in\mathrm{orb}(\mathrm{id}_{\mathsf{S}})$. So (H)(ii) holds. ☐

By Lemmas 3–4, the machinery of (T3) applies to $\mathcal{K}_\varepsilon$ for both ε: the class $[\omega(\mathcal{K}_\varepsilon)]\in H^2(\mathrm{Sk}_{\mathcal{C}};\mathcal{D})$ is defined and $(G)\Leftrightarrow[\omega]=0$.

4 The orbit category, the complex, and $H^2\cong\mathbb{Z}/2$

We compute the data of (T3) directly; everything below is analytic.

Orbit category $\mathrm{Sk}_{\mathcal{C}}$. Its objects are $\mathsf{S},\mathsf{W},\mathsf{R}$; its non-identity morphisms are the non-regular free orbits from Lemma 3:

$$[p]:\mathsf{S}\to\mathsf{W},\quad [q]:\mathsf{S}\to\mathsf{R},\quad [s],[s_2]:\mathsf{W}\to\mathsf{R}.$$

Composition is $\mathrm{orb}(c_2)*\mathrm{orb}(c_1)=\mathrm{orb}(c_2\circ c_1)$ (well defined by (T3) under (H)). The only composites of non-identities are

$$[s]*[p]=\mathrm{orb}(s\circ p)=\mathrm{orb}(q)=[q],\qquad [s_2]*[p]=\mathrm{orb}(s_2\circ p)=\mathrm{orb}(q\cdot\tau^\varepsilon)=\mathrm{orb}(q)=[q],$$

the last equality because q and $q\tau$ lie in the *same* orbit; hence $[s]*[p]=[s_2]*[p]=[q]$ *independently of ε*. So $\mathrm{Sk}_{\mathcal{C}}$ is the branch $\mathsf{S}\to\mathsf{W}\rightrightarrows\mathsf{R}$ with both middle arrows composing to $[q]$ — the same category for both ε.

Presheaf $\mathcal{D}$. $\mathcal{D}(\mathsf{S}) = \mathbb{Z}/2$, $\mathcal{D}(\mathsf{W}) = \mathcal{D}(\mathsf{R}) = 0$. Every restriction map $\varphi_c : G_{\operatorname{cod} c} \to G_{\operatorname{dom} c}$ along a generator has $\operatorname{cod} c \in \{\mathsf{W}, \mathsf{R}\}$ (the only generators are $[p], [q], [s], [s_2]$, none of which targets S), so $G_{\operatorname{cod} c} = 0$ and $\varphi_c = 0$: *all restriction maps vanish.*

Normalized cochain complex. By (T3), $C^n = \prod_{\text{composable } o_n,\dots,o_1} G_{\operatorname{dom} o_1}$ on non-identity chains, and only chains with $\operatorname{dom} o_1 = \mathsf{S}$ contribute (elsewhere $G_{\operatorname{dom} o_1} = 0$).

- C^1: generators with source S are $[p], [q]$, giving coordinates $h([p]), h([q]) \in \mathbb{Z}/2$; so $C^1 \cong (\mathbb{Z}/2)^2$.
- C^2: composable pairs (o_2, o_1) with $\operatorname{dom} o_1 = \mathsf{S}$. Then $o_1 \in \{[p], [q]\}$; if $o_1 = [q]$ then o_2 starts at R, a sink, so none; if $o_1 = [p]$ then $o_2 \in \{[s], [s_2]\}$. Coordinates $\omega([s], [p]), \omega([s_2], [p]) \in \mathbb{Z}/2$; so $C^2 \cong (\mathbb{Z}/2)^2$.
- $C^3 = 0$: a composable triple with $\operatorname{dom} o_1 = \mathsf{S}$ would need $o_1 = [p]$, $o_2 \in \{[s], [s_2]\}$, and o_3 starting at R — none exist.

Cohomology. Since $C^3 = 0$, $Z^2 = \ker \delta^2 = C^2 \cong (\mathbb{Z}/2)^2$. The coboundary δ^1 (with all $\varphi = 0$) is, from (T3) Eq. (d1),

$$(\delta^1 h)([s], [p]) = -h([s] * [p]) + h([p]) = h([p]) - h([q]), \qquad (\delta^1 h)([s_2], [p]) = h([p]) - h([q]),$$

so $B^2 = \operatorname{im} \delta^1 = \{(t, t) : t \in \mathbb{Z}/2\} = \langle (1, 1) \rangle$, the diagonal. Therefore

$$H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D}) = (\mathbb{Z}/2)^2 / \langle (1, 1) \rangle \cong \mathbb{Z}/2.$$

5 The obstruction class equals the token-mutation bit

Theorem 5 (Main)**.** *Let $T = \{p, q, s, s_2\}$ (together with identities) be the natural transversal of $\mathcal{K}_\varepsilon$. Its defect 2-cocycle is*

$$\omega_T = \big(\omega_T([s], [p]),\ \omega_T([s_2], [p])\big) = (0, \varepsilon) \in (\mathbb{Z}/2)^2,$$

and consequently

$$[\omega(\mathcal{K}_\varepsilon)] \ = \ \varepsilon \cdot (\textit{generator}) \qquad \textit{in} \ \ H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \mathbb{Z}/2.$$

Proof. The defect at a composable pair (c_2, c_1) of chosen generators is the unique $\mathcal{D}$-element $\omega_T(c_2, c_1)$ with $c_2 \circ c_1 = \lfloor c_2 c_1 \rfloor_T \circ \omega_T(c_2, c_1)$, where $\lfloor \cdot \rfloor_T$ is the chosen generator of the target orbit (T3, Def. of the defect). Both composable pairs have target orbit $\operatorname{orb}(q)$ with chosen generator q. Then:

$$s \circ p = q = q \circ \mathrm{id}_{\mathsf{S}} \ \Rightarrow \ \omega_T([s], [p]) = \mathrm{id}_{\mathsf{S}} \ (= 0);$$

$$s_2 \circ p = q \cdot \tau^\varepsilon = q \circ \tau^\varepsilon \ \Rightarrow \ \omega_T([s_2], [p]) = \tau^\varepsilon \ (= \varepsilon).$$

So $\omega_T = (0, \varepsilon)$. Its class in $H^2 = (\mathbb{Z}/2)^2 / \langle (1, 1) \rangle$ is $(0, \varepsilon)$; for $\varepsilon = 0$ this is 0, and for $\varepsilon = 1$ we have $(0, 1) \notin \langle (1, 1) \rangle = \{(0, 0), (1, 1)\}$, the nonzero generator. Hence $[\omega(\mathcal{K}_\varepsilon)] = \varepsilon \cdot (\text{generator})$. $\square$

Corollary 6 (Re-entrancy dichotomy)**.** *For the supervisor–worker orchestration $\mathcal{K}_\varepsilon$ with token-internal right factor $\mathcal{D}$:*

(i) *$\mathcal{K}_\varepsilon$ admits a Zappa–Szép product $\mathcal{K}_\varepsilon = \mathcal{C} \bowtie \mathcal{D}$ — equivalently a distributive law $\lambda : \mathcal{DC} \Rightarrow \mathcal{CD}$ (ZS1–ZS4), equivalently a serialization of the supervisor's dispatch and the worker's internal step into a single joint agent* — if and only if $\varepsilon = 0$, i.e. iff the worker's outcome fixes the supervisor's turn token *($s_2 \circ p = q$).*

(ii) *When $\varepsilon = 1$ (the worker mutates the token, $s_2 \circ p = q\tau$) the obstruction class $[\omega]$ is the nonzero generator of $H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \mathbb{Z}/2$; no distributive law exists. This is exactly the unprotected-re-entrancy failure mode.*

Proof. *(i)* By (L) (Lemma 3) and (T2), $\mathcal{C} \bowtie \mathcal{D}$ exists iff (G) holds. By (H) (Lemma 4) and (T3), (G) holds iff $[\omega] = 0$. By Theorem 5, $[\omega] = 0$ iff $\varepsilon = 0$. Chaining the three equivalences gives (i). *(ii)* is the case $\varepsilon = 1$ of Theorem 5 together with the same equivalence chain: $[\omega] \neq 0 \Rightarrow (G)$ fails $\Rightarrow$ no $\mathcal{C} \bowtie \mathcal{D}$. □

Remark 7 (Why the class *equals* the bit)**.** The two coordinates of ω_T are the two worker outcomes s, s_2; the coboundary subgroup $B^2 = \langle(1, 1)\rangle$ is the freedom to relabel which token-state is "canonical," which shifts *both* outcomes together. The class is nonzero precisely when the two outcomes *disagree* on that relabelling by one token-flip — and that disagreement is, by definition, the re-entrant branch's side effect $s_2 \circ p = q\tau \neq q$. Formally the coordinate difference $\omega_T([s_2], [p]) - \omega_T([s], [p]) = \varepsilon$ is the gauge-invariant content, and it is the token-mutation bit. This is the categorical avatar of "a flat bundle is trivial iff its holonomy vanishes": the local completions of each partial handoff (free, by (L)) glue to a global serialized agent iff a single $\mathbb{Z}/2$ monodromy — the token flip — vanishes.

Remark 8 (Relation to the rigid twist)**.** $\mathcal{K}_1 = \mathcal{K}_{\mathrm{bug}}$ is isomorphic, as a category with distinguished wide subcategory $\mathcal{D}$, to the ten-morphism rigid twist of [2] (send $\mathsf{S}, \mathsf{W}, \mathsf{R} \mapsto a, x, y$ and $\tau \mapsto g$). The present proof does *not* route through that isomorphism: it applies (T3) to $\mathcal{K}_\varepsilon$ directly and computes the class from the defect, so the rigid-twist identification is now a corollary (both compute the generator of the same $H^2 \cong \mathbb{Z}/2$) rather than a load-bearing machine check. The gain is the uniform statement $[\omega(\mathcal{K}_\varepsilon)] = \varepsilon$ across the parameter, which the single rigid-twist instance did not provide.

6 The composing regimes (context)

Corollary 6 is the deliverable. Two further regimes, already computed in [3] and *not* re-established here, complete the qualitative picture of "composition = Zappa–Szép product":

- *Independent supervisors* (read-only workers): the distributive law is the swap, $\mathcal{K} = \mathcal{C} \times \mathcal{D}$ (trivial matched pair).
- *Coherent non-abelian interleaving*: two supervisors over a shared worker pool give the dihedral matched pair $\mathcal{K} = \mathbb{Z}/3 \bowtie \mathbb{Z}/2 = S_3$; ZS1–ZS4 hold and the join is genuinely non-abelian — so the criterion is non-vacuous, it produces legitimate joint agents that are neither products nor obstructed.

These are illustrative (the two-supervisor reading is schematic) and remain at grade COMPUTED; the PROVED content of this note is Corollary 6.

7 Status and grant relevance

What is proved. Corollary 6: the analytic dichotomy "$\mathcal{C} \bowtie \mathcal{D}$ exists $\iff$ the worker fixes the token," with the obstruction identified as the explicit class $[\omega] = \varepsilon \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \mathbb{Z}/2$. The scaffolding (T2), (T3) is cited and proved elsewhere [1, 2, 4, 5]; the freeness (L), the hypothesis (H), the orbit category, the complex, the defect, and the class are all computed by hand (§3–5), with no enumeration of factorization systems and no machine-verified isomorphism. A parametrized script (`projects/scratch/orchestration_zs_parametrized.py`) confirms every analytic claim ($\mathcal{K}_\varepsilon$ is a category, (L) holds, $\mathrm{Sk}_{\mathcal{C}}$ is ε-independent, B^2 is the diagonal, $\omega_T = (0, \varepsilon)$, and — as an independent cross-check, not part of the proof — $\#\mathrm{SFS} = 2, 0$ for $\varepsilon = 0, 1$).

What is not claimed. No new cohomology (the H^2 machinery is Baues–Wirsching / Rosebrugh–Wood). No claim that a named production framework *is* $\mathcal{K}_\varepsilon$: the model is a minimal faithful abstraction of the two-worker re-dispatch pattern, empirically instantiable later. The novelty is the Zappa–Szép obstruction layer itself: the closest prior art (Aberlé, arXiv:2604.01303) supplies the polynomial-interface / free-monad implementation mechanism but only the *unobstructed* parallel sum of two systems — no Zappa–Szép product, no interaction obstruction, no cohomology. This theorem is that surviving layer, and it is the anchor for the grant's Impact section: *unprotected re-entrancy in agent orchestration is a nonzero degree-two class on the handoff category*, one degree above the H^0/H^1 sheaf-Laplacian invariants of the multi-agent-systems literature (which measure consensus and identifiability on the *communication* graph, not composability of the *handoff* structure).

References

[1] MacBeth, *The pairwise Zappa–Szép criterion*, Kodamai note, 10 June 2026.

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